

Elastic Bounds of Pearson's r : Theoretical and Empirical Insights for Ordinal Variables

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(Hardy et al., 1952)

1. Preamble

1.1 The ‘Point’:

The widespread use of Pearson’s r with ordinal data.

1. *Alert* applied researchers of ‘elastic bounds’
 1. Problems of ignoring the above $\leadsto$ errors in interpretation (e.g. Cohen and Cohen) – how max/min bounds of r vary with marginals.
2. *Provide* tools to assess if problems exist in own data/ research
3. *Clarify* that traditional hypothesis testing (e.g. t-test of $H_0: r = 0$) are incorrect and provide correct hypo testing (permutation/ randomization tests)

1.2 Contributions:

1. Theory of optimal bounds\
2. Symmetry-breaking conditions\
3. Empirical consequences $\leadsto$ hypo testing and significance
4. Supporting tools

1.3 Possible Literature

Grether, D. M. (1976). On the Use of Ordinal Data in Correlation Analysis. *American Sociological Review*, 41(5), 908–912. <https://doi.org/10.2307/2094741>

Labovitz, S. (1967). Some observations on measurement and statistics. *Social Forces*, 46(2), 151–160. <https://doi.org/10.2307/2574595>

Labovitz, S. (1970). The assignment of numbers to rank order categories. *American Sociological Review*, 35(3), 515–524. <https://doi.org/10.2307/2092993>

Moss, J., & Grønneberg, S. (2023). Partial Identification of Latent Correlations with Ordinal Data. *Psychometrika*, 88(1), 241–252. <https://doi.org/10.1007/s11336-022-09898-y>

O’Brien, R. M. (1979). The use of Pearson’s r with ordinal data. *American Sociological Review*, 44(5), 851–857. <https://doi.org/10.2307/2094532>

Robitzsch, A. (2022). On the Bias in Confirmatory Factor Analysis When Treating Discrete Variables as Ordinal Instead of Continuous. *Axioms*, 11(4), Article 4. <https://doi.org/10.3390/axioms11040162>

2. Introduction and Conceptual Setting

In this article we elaborate why correlations involving ordered-categorical variables are generally ‘constrained’, which means that their attainable maximum and minimum values

are rarely +1 and -1, as is often thought, but values that have to be calculated for each pair of variables. We assess the magnitude of these constraints in fictitious as well as empirical data and analyse under which conditions they are most severe. The article is accompanied by a software tool in R: that allows applied researchers to easily assess the extent of the problem in their own data.

The use of product-moment correlations (also known as Pearson correlations) to assess linear relationships between ordered-categorical variables –henceforth referred to as *orcats*– is widespread and widely condoned in empirical studies, in spite of such variables not being metric while the product-moment correlation assumes that they are. We do not focus on this problem from an obsessive desire for methodological purity, were it only because many statistical procedures are quite robust for violations of their assumptions [fn: examples?]. Instead, we draw attention to problematic consequences of this violation of assumptions that, as far as we know, have been rarely recognised in the applied literature although they are well-known in the theoretical statistical literature. This is the problem that correlation coefficients involving *orcats* are constrained in their range which makes them mutually incomparable, leading to problems in hypothesis testing and in analytical procedures that use correlations as their input.

This article proceeds as follows. We start by briefly elaborating the character of ordinal categorical variables (*orcats*), why they are generally not regarded as being metric, and why, in spite of that, they are nevertheless very frequently used in statistical procedures that assume metric level data. We then proceed in section 3 to show why correlations based on *orcats* tend to be constrained, which implies that their range is smaller than from -1 to +1, and why this is problematic in applied research. This raises the question how to determine the actual range of attainable values of correlations between *orcats*, which we address in the subsequent section. In section 5 we assess the magnitude of these constraining effects on two different sets of data. One is a typical survey dataset in which many of the variables are ordinal categorical, the *orcatorcat* 2019 British Election Study. The second consists of data derived from simulations covering all possible relevant conditions for pairs of ordered categorical variables. We use both sets of data to explore conditions that influence the magnitude of the constraining effect. Section 6 concludes by summarizing the findings, by elaborating the consequences for applied research, and by discussing directions for further research.

2.1 On ordinal-categorical variables (*orcats*), their level of measurement and their common use in quantitative analyses

Orcats are ordinal variables with a relatively small number of categories (or ranks) compared to the number of cases, thus resulting in partial orderings of cases. Although they can be found in all field of quantitative inquiry they are ubiquitous in data from standardised social science surveys. Likert items and rating scales [describe in fn Likert items and rating scales, add that there are yet other formats such as semantic-differential scales that also yield *orcats*] are among the most favoured ways to elicit responses in such surveys. These survey-question formats have most often between 3 and 11 ordinal

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categories. The numerical values assigned to the categories of *orcats* are most often consecutive integers, starting with either 0 or 1. [fn: explain that these values may be included in the formulation of a survey question, which is immaterial to our discussion here. The values assigned to the categories (and to the responses) reflect, in combination with the analysis procedure the analyst's interpretation of their meaning.], Such an assignment of values is as good as, and indeed analytically equivalent to, any other set of values that reflects the analyst's understanding of the ordinal relations between the categories, i.e., any monotonic transformation. However, objections are often raised when *orcats* [fn: add references] are analysed with procedures requiring metric measurement. That requirement implies that the differences between values can validly be compared so that, e.g., the substantive difference between categories 1 and 2 is equal to that between categories 3 and 4, and half as large as the difference between categories 2 and 4, etc. Any implied interval-level interpretation of ordinal categorical variables requires therefore a compelling theoretical or empirical justification of the numerical values assigned to the categories rather than just the analyst's fiat. In principle this is quite possible, for example by calibrating values of categories against psychophysical measurements (cf., Lodge; Tillie; others), or against the results of psychometric/linguistic analysis of verbal response labels (cf Sönning 2024). However, such procedures are rarely integrated in applied survey research because of their high cost. In the absence of compelling justifications for values assigned to categories *orcats* can only be considered to provide ordinal information, not metric information. In spite of this we find that they are very commonly analysed with statistical tools that require metric measurement of variables such as product-moment correlations or procedures that build upon such correlations. We find this clearly exhibited in published articles in high-ranking academic journals [fn: explain how we did this] and in highly regarded textbooks where examples are given of the use of correlations between *orcats* as input for procedures such as PCA, FA or SEM [fn: examples: Kline; Byrne; xxx]. This practice is so established that it is occasionally rationalised by referring to *orcats* as being 'quasi-interval level' measures [fn: reference]. One can, however, also find more explicit justifications for this practice, which emphasise the robustness of correlation coefficients for violations of their measurement level assumptions. Labovitz (1967, 1970) in particular has compared correlations between *orcats* scored according to equal-distances with correlations when the variables were scored with a monotonic random procedure (which only assumes ordinality between categories). He reports that the differences are negligible, and suggests that even if the theoretical objection against unfounded interval level interpretations of *orcats* would be justified, that objection would nevertheless have such minor practical consequences as to make it irrelevant in most situations. Not surprisingly, Labovitz' work has in turn be criticised itself on various grounds, most particularly of (inadvertently) mainly covering situations where this robustness holds, and overlooking other situations (cf O'Brien 1979; xxx). At this place we will not focus on discussions about the equal-distance scoring practice. Our main reason for referring to them is to demonstrate the existence of lively debates in the applied literature about potential issues when using *orcats* in procedures requiring metric data. The existence of these debates makes it remarkable that another important issue has – to the best of our knowledge – remained mainly overlooked in the applied literature while it

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has been well-known in the theoretical statistical literature. This is the problem that correlations between [orcats](#) are generally ‘constrained’, which means that they do not range between -1 and +1, but between minima and maxima that have to be calculated separately for each pair of [variables](#). The nature of this issue is the subject of the next section.

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2.2 Bounds of correlations between ordinal categorical variables ([orcats](#))

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The values that product-moment correlations between ordinal-categorical variables ([orcats](#)) can attain are generally constrained by minimum and maximum bounds of attainable values that are not -1 and +1. [fn: The maximum attainable value is +1 only when the univariate distributions of the variables are identical. But that does not imply the minimum attainable value then to be -1, as that also requires further conditions that the univariate distributions must meet (see below, in the discussion of Table 2). The minimum attainable value of a correlation is -1 when the univariate distributions are each other’s inverse, but again, that does not imply that the maximum is +1, as that requires additional conditions of the univariate distributions.]

Because the bounds of correlations between [orcats](#) are not -1 and +1 it is even more difficult than commonly thought in applied research to interpret the strength of correlations. The well-known and often referred to criteria formulated by [Cohen \(1988: 77-81\)](#) for correlations in the behavioural sciences [fn: Cohen suggested, with appropriate caution, the following guidelines: 0.10<r<0.29 weak; 0.30<r<0.49 moderate; 0.50<r<1.00 strong. [Hemphill \(2003\)](#) suggests on the basis of a review of several meta-analyses in psychological research that the cut-off of 0.50 for considering a correlation to be ‘strong’ may be too high in many common instances.] are explicitly based on the notion that the minimum and maximum attainable values for r are -1 and +1. But that is generally not the case and, moreover, the actual upper- and lower-bounds of r are not the same across pairs of variables, but specific to each. This state of affairs thus generates two interconnected problems when correlating [orcats](#). First, the absence of clarity of the lower- and upper-bounds of r makes qualitative interpretations of the strength of correlations elusive. Second, because the bounds of attainable values of r are specific for each pair of variables each correlation is expressed in terms of a different, but unknown reference interval. Such correlations are therefore not comparable. To better grasp these problems, it is necessary to determine the upper- and lower bounds of correlations, given the univariate distributions of the variables involved.

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Statistical scholarship has since long recognized that, in contrast to what is often thought, the upper- and lower-bounds of correlations between [orcats](#) are generally not the reference interval [-1, +1]. [insert references, including Shi&Huang 1992: xxxx]. Determining the actual bounds given the marginal distributions of the two variables can be done by using [Whitt’s \(1976\)](#) rearrangement theorem and [Hardy–Littlewood–Pólya’s \(1952\)](#) rearrangement inequality. For obtaining the maximum attainable value of a correlation the values of X and Y have each to be sorted independently in either ascending or descending order and then paired (comonotonic arrangement). For the minimum attainable value of

the correlation (given the marginals of the two variables) the anti-comonotonic arrangement is required (sorting one of the variables in ascending and the other in descending order). A formalized elaboration is presented in Appendix 1. Moreover, the algorithm for calculating r_{min} and r_{max} has been programmed in R and is [or: will be] freely available for researchers to assess the minimum and maximum attainable values of r for pairs of variables in their own data. Details about this routine are described in Appendix 2.

Table 1 presents an illustrative example of comonotonic and anti-comonotonic arrangements and of the resulting lower- and upper-bounds of r , for two 4-category variables and 12 cases. One can easily expand this example to pairs of variables with different numbers of categories and different numbers of cases.

A series of illustrative cases is presented in Table 2 (as in Table 1 for $n=12$ and two orcats with 4 categories each). Although the distributions of the variables given in these examples is of no particular importance, and could have been replaced by innumerable other examples, they are nevertheless helpful to arrive at some general observations about r_{min} and r_{max} :

- The values of r_{min} and r_{max} are not constant across all examples, and not even across correlations involving a specific variable (see examples (a) and (b) which involve the same X-variable). The variation in the values of the bounds of r is across pairs of variables. In other words, the bounds of r are specific for each pair.
- r_{min} can be -1, and r_{max} can be 1, but neither of these values implies the other (as seen in examples (b) and (c). The combination of $r_{min}=-1$ and $r_{max}=1$ is possible, but dependent on specific characteristics of the two marginal distributions involved (see examples (d) and (f). Both variables must not only have identical distributions, but these must also be symmetrical.
- r_{min} and r_{max} are not necessarily symmetric around 0. Such symmetry may exist, but requires specific characteristics of the pair of marginal distributions, namely that both are symmetric (see examples (e), (g) and (h).
- The extent to which the reference interval $[-1, 1]$ is covered by the interval $[r_{min}, r_{max}]$ varies across pairs of variables.
- The values of r_{min} and r_{max} depend not only on the marginal distributions of the two variables, but also on the values assigned to their categories. Unless specified differently, we follow in this article the common practice of assigning consecutive integer values to the ordered categories. But any monotonic transformation of values is equally adequate to represent the ordinal character of variables. Examples (i) and (j) involve the same pair of variables, but in (i) their categories have been assigned values 1, 2, 3 and 4, while in (j) the assigned values were 1, 2.25, 2.75 and 4. The consequences of this change are visible in the values of r_{min} and r_{max} . This observation implies that value assignment to categories of orcats (and the justification thereof) remains an issue of importance when using Pearson

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correlations on orcats. It is a different problem, however, than that of constraintment of correlations on which we focus here.

2.3 The magnitude [strength?] of marginal constraints on the bounds of r

As demonstrated above, the extent to which marginal distributions of orcats constrain the range of attainable value of r varies considerably. In the few illustrative examples presented in Table 2 rmin varied between -1 and -0.71, while rmax varied between 0.32 and 1. Moreover, in Table 2 rmin and rmax do not vary in lockstep but at least to some extent independent of each other.[fn: Table 2 is a poor basis to assess the functional relationship between rmin and rmax as it contains a small number of handpicked situations. This relationship will be investigated in detail later, in section ??.] This raises the question how strong the overall constraining effect is that marginal distributions exert on the attainable values of r. It also raises the question of what factors affect the strength of this overall constraining effect. In this section we describe two ways in which the strength of constraints can be defined. Later (in section ??) we turn to the questions of the conditions that drive the strength of this effect.

Perhaps the most obvious way to express the strength of the marginal constraints on the attainable values of r is by comparing the width of the interval between rmin and rmax with that of the ubiquitously used reference interval [-1, 1]. As the width of the latter interval is 2, it is advantageous to express the strength of the overall constraining effect—to be denoted as C_1 as

$$C_1 = \frac{|r_{max}| + |r_{min}|}{2}$$

C_1 reflects the proportion of the range of the attainable values of r compared to the theoretically maximum width of this range.

Applying this measure to the illustrative cases listed in Table 2 shows that C_1 varies. Its highest value is 1, reflecting the absence of any constraining effect (see examples (d) and (f) in Table 2). The lowest value of C_1 in Table 2 is for example (i), where it is 0.639. In other words, the marginal distributions of the two variables in that example restrict the range of its attainable values to 63.9% of the traditional [-1, 1] reference interval.

Some might object to the C_1 measure on the grounds that it assumes that r (and thus also rmin and rmax) is expressed at interval level and that one could thus state validly that a correlation of 0.4 reflects an association twice as strong as one of 0.2. That would be so if numerical differences between correlations correspond to similar differences in observable aspects of data [insert references to correspondence NRS to ERS]. That not being the case, one could instead compare r^2 s—percentages of shared variance—rather than rs. [fn: This perspective is not uncontested, however, and its usefulness depends on the purpose of inspecting correlations: as reflections of the association between variables, or as reflections of associations between indicators of a latent concept (for an elaboration

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of this difference, cf. [Ozer \(1985\)](#).] This would lead to a somewhat different measure of the strength of the constraining effect of marginal distributions on the attainable values of r , as follows:

$$C_2 = \frac{r_{max}^2 + r_{min}^2}{2}$$

For the examples in Table 2, the values C_2 are, respectively, 0.701; 0.755; 0.715; 1; 0.831; 1; 0.809; 0.915 and 0.510. Obviously, C_1 and C_2 are correlated, but how strongly cannot be assessed very well from the few handpicked examples in Table 2, but it will be considered below, using more encompassing sets of data.

2.3 Data to be used for investigating the bounds of attainable values of r

To investigate factors that affect the bounds of attainable values of r and thus the strength of constraints on r exerted by marginal distributions we use two different sets of data.

Our first dataset is simulation generated and covers in principle all possible combinations of marginal distributions for two orcats with given numbers of categories and given n . We simulate orcats containing, respectively, 4, 5, 6, 7, 10 and 11 categories, these being by far the most common in empirical data deriving from survey studies. For each of these we generate a large number of variables, each with its own simulated univariate distribution, and thus also its own form (modality, symmetry, etc.), mean, variance, skew and kurtosis as follows. [insert description, detailing n , how many univariate distributions we generate for each number of categories; how to populate the categories, whether we impose distributional characteristics such as normality, modality, skew etc. Btw: my preference would be to impose as little as possible, and derive characteristics such as mean, variance, skew, kurtosis from the distributions generated.] We subsequently randomly select pairs of simulated variables and determine for each r_{min} , r_{max} , C_1 and C_2 . From the resulting simulated pairs of orcats we delete the cases in which one, or both of the variables have zero variance. This results in a dataset with $n = xxx$ [insert very large number] that forms the basis of subsequent analyses. Further details are specified in Appendix 3.

This simulated dataset constitutes, within the bounds of granularity of the procedure, a random sample all possible combinations of univariate distributions of pairs of orcats, and can therefore be used as an adequate basis for investigating functional relationships between sundry properties of pairs of orcats and the range of attainable values of r for those pairs.

Although this simulated dataset randomly represents all possible combinations of univariate distributions of pairs of orcats, it cannot be considered to be a random sample of pairs of orcats that applied researchers would encounter in actual, empirical data such as derived from, e.g., mass surveys. This is because, when designing survey questions, researchers tend to avoid formulations that they expect to yield little variance in the responses. Therefore, we use a second dataset to investigate constraints on the attainable

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range of values of r , that is derived from the British Election Study 2019. [fn: insert details about BES2019]. In this study we identified xxx orcats, which define 7503 pairs, for each of which we calculated pair-wise distributional characteristics including r_{min} , r_{max} , C1 and C2. Further details on this dataset are provided in Appendix 4.

Obviously, the orcats in BES2019 are not a random sample from the contents of all socio-political mass surveys in Britain in 2019, let alone of those in wider areas such as the developed western world, or in other years. But the BES can sensibly be regarded as *typical* for such studies, which include (national) election studies, general social surveys, and many public opinion and marketing surveys. As such, findings based on this dataset derived from the BES can sensibly be regarded as typical for what applied researchers in other countries will encounter in their survey data.

These two datasets will be used to address, in the subsequent sections, questions with regard to:

- Distributions of r_{min} , r_{max} and measures of overall constraint of r by marginals
- Factors affecting r_{min} , r_{max} and overall constraint of r
- Consequences of constrained correlations in applied research, including for descriptive purposes and hypothesis testing
- Possibilities for reducing undesired consequences of constrained correlations

2.3 Background and Definitions

- Ordinal data and assumptions of interval-scale correlation
- Existing alternatives: Spearman, polychoric, polyserial; their strengths and limitations
- Coupling and rearrangement in discrete settings

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 <#>Conclusion: for every given distribution of a variable, there are only two distributions of the other variable which do not constrain r : the same, and the inverse. Every other distribution does constrain r ¶
 <#>In case of constraint ('shrinkage?'), the relationship becomes partially nonlinear, by necessity (does this mean that the problem of constraint does not exist for eta?) Is there a measure for nonlinearity? (polynomial multiple correlation?)¶
 Determining the minimum and maximum attainable values of r for given marginal distributions¶
 <#>Explain different strategies which will all yield the same answers: separate sorting and pairing / Frechet-Hoeffding bounds/ etc¶
 <#>Proofs in appendix?¶
 How severely are correlations constrained – or: how long is a piece of string?¶
 <#>Explain strategy of using BES data and resulting dataset of 7503 item-pairs¶
 <#>Descriptive analyses:¶
 <#>Define a measure for extent of constraint¶
 <#>Describe univariate distribution of this measure; asymmetry etc. -Probability of reaching the r_{max} and r_{min} (and implications: does this mean that the constraint is even more severe?)¶
 Factors influencing the degree to which correlations are constrained¶
 <#>Analyses of the same dataset of 7503 item pairs¶
 <#>Other characteristics of these include¶
 <#>Characteristics of each of the two vars separately (their variance / skew / kurtosis / mean etc), as well as dyadic characteristics (difference of means / skew/ etc)¶
 <#>Hypotheses/expectations:¶
 <#>Effect of difference of means and of difference of skew¶
 <#>Perhaps effect of number of categories (constraint less severe the larger the number of categories)¶
 Consequences of constrained correlations in applied research¶
 <#>Descriptive interpretation of correlations (Cohen's criteria) is undermined¶
 <#>It is also undermined by every correlation being expressed on its own unique scale/unique calibration¶
 <#>Need a nice example of 2 correlations with $r(a, b) \leq r(b, c)$ and where $r(a, b)_{max} < r(b, c)$: is it smaller because the linear relationship is weaker, or because the constraint is stronger?¶

3. Theoretical Framework

3.1 Optimal Coupling and Correlation Bounds

We formalize the problem of identifying the joint distribution (coupling) of two ordinal variables with fixed marginals that maximizes or minimizes the Pearson correlation.

Proof. The Pearson correlation coefficient is given by:

$$r_{XY} = \frac{E[XY] - E[X]E[Y]}{\sigma_X \sigma_Y}.$$

Given fixed marginals $p_X(x_i)$, $p_Y(y_j)$, the expectations $E[X]$, $E[Y]$ and standard deviations σ_X , σ_Y are constant. Thus, to maximize or minimize r_{XY} , we only need to maximize or minimize the expectation $E[XY]$:

$$E[XY] = \sum_{i=1}^{K_X} \sum_{j=1}^{K_Y} x_i y_j P(X = x_i, Y = y_j).$$

This proof relies on a rearrangement inequality of (Hardy et al., 1952) (Theorem 368):

For real sequences $a_1 \leq a_2 \leq \dots \leq a_n$ and $b_1 \leq b_2 \leq \dots \leq b_n$, and for any permutation π of indices $\{1, \dots, n\}$ we have the following:

$$a_1 b_1 + a_2 b_2 + \dots + a_n b_n \geq a_1 b_{\pi(1)} + a_2 b_{\pi(2)} + \dots + a_n b_{\pi(n)}$$

and

$$a_1 b_n + a_2 b_{n-1} + \dots + a_n b_1 \leq a_1 b_{\pi(1)} + a_2 b_{\pi(2)} + \dots + a_n b_{\pi(n)}$$

Equality occurs only if the permutation π results in a monotonically increasing sequence (for maximum) or monotonically decreasing sequence (for minimum).

To explicitly connect this to our problem, enumerate observations explicitly from sorted marginals:

$$X_1: x_{[K_X]} \geq x_{[K_X-1]} \geq \dots \geq x_{[1]}, \quad Y_1: y_{[K_Y]} \geq y_{[K_Y-1]} \geq \dots \geq y_{[1]}.$$

(Sorted in descending order for maximization). Next expand discrete probability distributions into explicit observation-level sequences. If $K_X \neq K_Y$, extend the shorter sequence with zero-probability categories so both sequences have equal length, which does not affect marginal probabilities or correlation.

Hence, to maximize $E[XY]$: pair the largest ranks of X with the largest ranks of Y . Explicitly, start from the top (highest values) and move downward:

- Pair as many observations of the largest X -category as possible with the largest Y -category, then next-largest, etc., until all observations are paired.

- By the Hardy–Littlewood–Pólya Rearrangement Inequality (**hard52?**), this explicitly constructed pairing yields the maximum possible sum of product(Hardy et al., 1952)s, hence maximizing $E[XY]$.

Likewise, to minimize $E[XY]$ pair the largest X -categories with the smallest Y -categories, then second-largest X -categories with second-smallest Y -categories, and so forth (anti-comonotonic ordering). By the rearrangement inequality, this arrangement explicitly yields the minimal sum of products, hence minimizing $E[XY]$.

In words, given fixed marginals, we have shown:

- **Maximum correlation** $r_{\max}$ is achieved by comonotonic (e.g. sorting values of X and Y from largest to smallest independently and pairing) arrangement:

$$P_{\max}(X = x_{[i]}, Y = y_{[i]}) \quad \text{in either descending or ascending order.}$$

- **Minimum correlation** $r_{\min}$ is achieved by anti-comonotonic arrangement:

$$P_{\min}(X = x_{[i]}, Y = y_{[n+1-i]}) \quad \text{pairing descending } X \text{ with ascending } Y, \text{ or viceversa.}$$

With this explicit construction we can simply calculate $r_{\max}$ and $r_{\min}$ when the values of X and Y are ranks.

Proof. Let the total number of observations be N , thus:

$$\sum_{i=1}^{K_X} n_X(i) = N, \quad \sum_{j=1}^{K_Y} n_Y(j) = N.$$

Means and standard deviations clearly become:

Means:

$$\bar{X} = \frac{1}{N} \sum_{i=1}^{K_X} i \cdot n_X(i), \quad \bar{Y} = \frac{1}{N} \sum_{j=1}^{K_Y} j \cdot n_Y(j).$$

Standard deviations:

$$\sigma_X = \sqrt{\frac{1}{N} \sum_{i=1}^{K_X} (i - \bar{X})^2 n_X(i)}, \quad \sigma_Y = \sqrt{\frac{1}{N} \sum_{j=1}^{K_Y} (j - \bar{Y})^2 n_Y(j)}.$$

To achieve the maximum expected product $E[XY]_{\max}$, do the following: First sort ranks from largest to smallest independently for both X and Y ; then pair ranks directly from highest to lowest available observations.

The formula for $E[XY]_{\min}$ is derived in a similar manner. Define explicitly a matrix $R_{i,j}$ giving frequencies of pairings $X = x_i$ with $Y = y_j$. For the minimal expectation, pair largest available X -values explicitly with smallest available Y -values first (greedy algorithm):

Initialize explicitly available frequencies:

$$n'_Y(j) = n_Y(j) \quad \text{for each } j$$

- For i from largest to smallest ($i = K_X$ down to 1):
 - For j from smallest to largest ($j = 1$ up to K_Y):
 - Pair explicitly as many observations as possible:

$$R_{i,j} = \min \left(n_X(i) - \sum_{s=1}^{j-1} R_{i,s}, n'_Y(j) \right)$$

- Update explicitly available frequencies:

$$n'_Y(j) \leftarrow n'_Y(j) - R_{i,j}$$

Then explicitly the minimal expectation is:

$$E[XY]_{\min} = \frac{1}{N} \sum_{i=1}^{K_X} \sum_{j=1}^{K_Y} x_i y_j R_{i,j}$$

This explicit equation applies Whitt's rearrangement theorem, Whitt (1976), ensuring proper maximal pairing of ranks. (Whitt (1976))

Discussion

We want to construct a joint distribution table M such that:

- The **row sums** match the marginal counts for X : $\sum_j M_{i,j} = n_X(i)$
- The **column sums** match the marginal counts for Y : $\sum_i M_{i,j} = n_Y(j)$

To do this, we go **greedily**, pairing the highest available X values with the **smallest available** Y values, one cell at a time.

To avoid **exceeding** the available marginal totals in Y , we must track how much of $n_Y(j)$ is **still unassigned** at each step — and that's what $n'_Y(j)$ represents.

After assigning $M_{i,j}$ units to the cell (i,j) , we subtract that amount from the remaining pool of level y_j values, like this: $n'_Y(j) \leftarrow n'_Y(j) - M_{i,j}$. This ensures we don't **overfill** any column of the matrix.

To maximize the expectation $E[XY]$, the rearrangement inequality states we must pair values of X and Y from smallest-to-smallest, second-smallest-to-second-smallest, and so forth (comonotonic alignment).

Thus, explicitly:

- Sort X : lowest-to-highest:

$$X_{\text{sorted}} = \begin{pmatrix} x_1, \dots, x_1, x_2, \dots, x_2, \dots, x_{K_X}, \dots, x_{K_X} \\ \check{n_X(1)} \quad \check{n_X(2)} \quad \check{n_X(K_X)} \end{pmatrix}$$

Sorted Y -values (ascending order for max, descending for min):

$$Y_{\text{sorted(max)}} = \begin{pmatrix} y_1, \dots, y_1, y_2, \dots, y_2, \dots, y_{K_Y}, \dots, y_{K_Y} \\ \check{n_Y(1)} \quad \check{n_Y(2)} \quad \check{n_Y(K_Y)} \end{pmatrix}$$

$$Y_{\text{sorted(min)}} = \begin{pmatrix} y_{K_Y}, \dots, y_{K_Y}, y_{K_Y-1}, \dots, y_{K_Y-1}, \dots, y_1, \dots, y_1 \\ \check{n_Y(K_Y)} \quad \check{n_Y(K_Y-1)} \quad \check{n_Y(1)} \end{pmatrix}$$

The maximum and minimum expectations are simply:

Maximum expectation (pair element-by-element):

$$E[XY]_{\text{max}} = \frac{1}{N} \sum_{k=1}^N X_{\text{sorted}}(k) \cdot Y_{\text{sorted(max)}}(k)$$

- Minimum expectation:**

$$E[XY]_{\text{min}} = \frac{1}{N} \sum_{k=1}^N X_{\text{sorted}}(k) \cdot Y_{\text{sorted(min)}}(k)$$

These classical bounds define feasible joint distributions given marginals and constrain the attainable values of r . There is an interesting connection to Fréchet–Hoeffding and Boole–Fréchet Inequalities, which we note but leave for future research.¹

¹ Fréchet–Hoeffding and Boole–Fréchet Inequalities

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You can also include code blocks:

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4. Measurement Issues and Metrics of Constraint

IN THIS SECTION: “How to define metrics (on *pairs* of discrete probability distributions) to *explain*:

3.

$$\frac{r_{\max} - r_{\min}}{2}$$
$$1 - r_{\max}$$
$$|-1 - r_{\min}|$$

Or analogs for r^2 :

1a.

$$\frac{r_{\max}^2 - r_{\min}^2}{2}$$

3.

$$\frac{(r_{\max} - r_{\min})^2}{2}$$
$$1 - r_{\max}^2$$
$$1 - r_{\min}^2$$

4.1 Possible “Metrics of Constraint”

[TO DO: define these. Add others?] Measures of ‘asymmetry

Aitchison distance (Aitchison (1989))

Entropy(?)

Total Variation(?)

4.2 Symmetry and the Role of Ties

- Conditions under which $r_{\min} = -r_{\max}$
- How ties in marginal distributions affect the attainability of the lower bound

4.3 Special Cases and Boundary Conditions

- Equal vs. unequal numbers of categories
- Symmetric vs. asymmetric marginal distributions

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Or even more paragraphs and content as needed.

5. Monte Carlo Simulations

- What is the ‘universe of possible’ (data)?

5.1 Probabilistic modeling over the $r \in [r_{\min}, r_{\max}]$ interval (?)

5.2 Effects of Marginal Shapes

- Shape-based elasticity of correlation bounds
- Skewness and modality heavily influence the attainable range of r .
- Marginals with extreme concentration (e.g., most responses in one or two categories) sharply restrict the possible correlation range.

5.3 Symmetry Breaking in Correlation Bounds

- When and why $r_{\min} \neq -r_{\max}$
- Empirical indicators of bound asymmetry
- When given marginal distributions are sufficiently asymmetric, the min and max bounds of r are not symmetric about zero.
- The shape of the null distribution of r can be strongly skewed, especially in cases with unequal marginals.

When we talk about “asymmetry” in this context, we’re referring to how different the distribution of Y is from the distribution of X . Since X is kept uniform, any deviation of Y from uniformity creates asymmetry between the distributions.

The *total variation distance* (TV) quantifies this asymmetry:

- It measures how different Y is from being symmetric around its midpoint
- When TV = 0: Y is symmetric (like X)
- When TV is large: Y is highly asymmetric compared to X

6. Illustration with ‘Typical’ Data: BES

We do not claim this data is representative of data-sets of ordinal variables used by social scientists. Rather, we view it as *typical* in that it is a real-world example possessing many of the features that would raise issues...

7. Applications and Implications

Impacts standard methods:

- **PCA and Factor Analysis:** Attenuated correlations may lead to $\leadsto$
 - Practical risks: over-factoring (more factors estimated than ‘should be’), biased structural coefficients(?)
 - **SEM:** Biased estimation of structural paths if ordinal-level attenuation is ignored.

Recommendations for interpreting Pearson’s r in applied contexts

- Reporting observed r in context of its attainable bounds.
- Using our tools to test whether an observed r is unusually strong or weak given marginals.

8. Extensions and Open Questions

- Role of entropy and joint uncertainty in bounding behavior
- Possibility of adjusting or normalizing r
- Relation to Copula Theory (Optional)
 - “Although our setting is discrete, the problem is conceptually linked to copula-based modeling of dependence with fixed marginals. The connection is discussed as a direction for future work.”

8.1 To Do

- Can $r_{min} > 0$?
- Are there systematic biases in testing $H_0: r = 0$ using t-test? (Answer with Monte Carlo simulated data?)
- Add: “ r_{min} is invariant to ‘problem 1’ (value assignment. So using ranks is actually w.l.o.g.)

9. Conclusion

- Summary: Theoretical bounds, symmetry-breaking conditions, and implications for applied research
- An R package accompanies this paper, providing tools to compute max/min bounds and perform randomization-based hypothesis tests for fixed marginals. Details are available in the appendix and online.
- Future work: deeper integration with copula models, correction strategies, and generalizations to higher dimensions

10. References

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11. Appendix A: Mathematical Proofs

- Formal derivations of optimal coupling and analytical bounds

12. Appendix B: Software Tools and Code Listings

- Documentation and examples for the R package
- Description of randomization testing features

13. Appendix C: Additional Tables and Simulations

- Supplementary empirical examples, simulation results

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